1. In this question, we will construct the tangent plane to the graph of the function

$$f(x, y) = \sin(x + y)$$

at a particular point P.

(a) Let $P = (\pi/2, 0, c)$. For what value of c is P on the graph of f?

(b) In single variable calculus, how did we compute the equation of the tangent line to the graph of a function $g(x)$ at a point (x_0, y_0)?

(c) How can we obtain the graphs of two single variable functions that go through the point P?

(d) For each of these single variable functions, determine the slope of their tangent lines.

(e) Using these slopes, find a direction vector (in $\mathbb{R}^3$) for each of these tangent lines.

(f) Compute the parametric forms for each of these tangent lines.

(g) What is the relationship between these tangent lines and the tangent plane to the graph of $f(x, y)$?

(h) Use the tangent lines you computed to find a (scalar) equation for the tangent plane to the graph of f at the point P.

Takeaway: computing the tangent plane to the graph of a function $f(x,y)$ at a point $P = (x_0, y_0, f(x_0, y_0))$

We can use the tangent plane to obtain an approximation of the function $f(x,y)$ at points (x,y) close to (x_0, y_0):

$$L(x,y) =$$

This also allows us to estimate the change in the values of f for (x,y) close to (x_0, y_0):

Definition:

2. Estimate the volume of paint needed to cover a cylinder of height 5m and radius 1m with a coat of paint 0.01m thick. (Recall that the volume of a cylinder with radius r and height h is given by $\pi r^2 h$)